# BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2025

# CHEMISTRY

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

## Section 'A' (Multiple Choice Questions)

**Maximum Marks:** 17

Note: This section consists of 17 part questions and all are to be answered. Each question carries one mark. Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part. The use of scientific calculator is allowed. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 3 Hours — Maximum Marks: 85 (Science Pre-Engineering & Pre-Medical Groups)

Q1. Choose the correct answer for each from the given options:

Qi. At 4°C the density of water is 1 gm/cm$^3$ then the mass of 2 dm$^3$ of water is:

(A) 8gm
(B) 100gm
(C) 1000 gm
(D) 2000gm

Qii. Total number of atoms in one mole of methane ($CH_4$) is:

(A) $6.02 \times 10^{23}$
(B) $3.01 \times 10^{24}$
(C) $1.81 \times 10^{23}$
(D) $30.1 \times 10^{24}$

Qiii. This colour has longest wave length:

(A) Red
(B) Blue
(C) Green
(D) Orange

Qiv. The magnetic quantum number for the last sub-shell having 3 electrons in phosphorous $_{13}P^{31}$ is:

(A) -1, 0, +1
(B) -1, 0, -1
(C) -1, 0, +2
(D) -1, 0, -2

Qv. Maximum number of electrons in a sub-shell is:

(A) 2l + 1
(B) 2l - 1
(C) 2(2l + 1)
(D) 2(2l + 1)

Qvi. This is the least polar molecule:

(A) HF
(B) HCl
(C) HBr
(D) HI

Qvii. The rate of diffusion of gas is 1/2 of that hydrogen then its molecular mass will be:

(A) 8 amu
(B) 10 amu
(C) 25 amu
(D) 50 amu

Qviii. This pair of gases do not obey Dalton's law of partial pressure:

(A) $H_2$ and He
(B) $H_2$ and $O_2$
(C) $N_2$ and $O_2$
(D) $NH_3$ and HCl

Qix. This compound does not contain hydrogen bond:

(A) $CH_4$
(B) $H_2O$
(C) $NH_3$
(D) HF

Qx. In crystalline structure, the length of axes and angles are unequal then crystal system will be:

(A) Cubic
(B) Monoclinic
(C) Orthorhombic
(D) Triclinic

Qxi. Kc = Kp, when Δn is equal to:

(A) zero
(B) -1
(C) +1
(D) -2

Qxii. This pair of ions is not iso-electronic:

(A) $Na^+$ and $Mg^{2+}$
(B) $Na^+$ and $N^{-3}$
(C) $Na^+$ and $S^{-2}$
(D) $Al^{3+}$ and $Na^+$

Qxiii. Quantum number values for 3p orbital are:

(A) n = 2, l = 1
(B) n = 3, l = 2
(C) n = 3, l = 0
(D) n = 3, l = 1

Qxiv. Two different solid having same crystal structure are called:

(A) Allotropy
(B) Polymorphous
(C) Isotopes
(D) Isomorphous

Qxv. A colloidal solution of a liquid into another liquid is called:

(A) Emulsion (B) Foam

(C) Gel (D) Sol

Qxvi. This liquid shows maximum surface tension:

(A) Water (B) Ethyl Alcohol

(C) Mercury (D) Gasoline

Qxvii. A reaction is first order with respect to "A" and second order with respect to "B" then the rate expression is:

(A) Rate = $K[A][B]^2$ (B) Rate = $K[A]^2[B]$

(C) Rate = $K[A][B]$ (D) Rate = $K[A][B]^3$

## Section 'B' (Short-Answer Questions)

**Maximum Marks:** 36

> Note: Answer any NINE part questions. All questions carry equal marks.

Q2.

(i) Define any four of the following: ☆ Significant Figure ☆ Rounding Off Data ☆ Bond Energy ☆ Dipole Moment ☆ Stoichiometry

(ii) Calculate the volume of $CO_2$ at 25°C and 780 torr pressure when 12.45 grams of marble is heated: $CaCO_3 \rightarrow CaO + CO_2$. —OR— 200 grams of $K_2Cr_2O_7$ reacted with 200 grams of concentrated $H_2SO_4$. Calculate: ☆ Mass of atomic oxygen produced ☆ Mass of reactant left unreacted. $K_2Cr_2O_7 + 4H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3[O]$

(iii) State Pauli's Exclusion principle. Write the electronic configuration of the following: ☆ $Ca^{++}$(Z = 20) ☆ $Br^{-}$(Z = 35) ☆ Ag(Z = 47)

(iv) Give scientific reasons of any four of the following: ☆ A drop of ink spreads on a blotting paper. ☆ Glycerin is distilled at reduced pressure. ☆ Mercury has its meniscus upward. ☆ HF are greater ionic character then HCl. ☆ Powered zinc reacts more rapidly with HCl than the chunks of metallic zinc.

(v) What is ideal gas? Drive the general gas equation. —OR— Write the postulates of Kinetic Molecular Theory of gases.

(vi) One mole of $NH_3$ gas occupies the volume of one $dm^3$ at 25°C, assuming $NH_3$ as a real gas, determine its pressure, while: a = 4.170 $dm^6$ atm $mol^{-2}$, b = 0.0371 $dm^3$/mol

(vii) State Le-Chatelier's Principle in Haber's process: $N_2 + 3H_2 \rightleftharpoons 2NH_3$ ($\Delta H$ = -ve)

(viii) Explain Lewis Theory of acids and bases with example.

(ix) State Raoult's Law and drive its mathematical expression.

(x) A glucose solution contains 0.15 mole of Glucose and 5.8 moles water at 20°C. Determine the lowing in the vapour pressure if vapour pressure of pure solvent at 20°C is 17.5 torr (assume solution is an ideal)

(xi) Explain the effect of the following factors on rate of a chemical reaction. ☆ Concentration of reactant ☆ Temperature

(xii) Explain Buffer solution and write its applications.

(xiii) If 255cm³ oxygen gas is collected over water at 25°C and a total pressure of 785 torr. Than find the mass of dry oxygen gas (Pressure of water vapour at 25°C is 24 torr)

(xiv) Draw molecular orbital diagram of $O_2$ molecules. Write molecules orbital configuration and determine its bond order.

## Section 'C' (Detailed Answer Questions)

**Maximum Marks:** 32

Note: Attempt any Two questions from this section. All questions carry equal marks.

Q3.

(a) what are X-rays? How are they produced? Give any four properties of X-rays also derive an expression for the energy of electrons in $n^{th}$ orbit according to Bohr's Model of atom when: $r = \varepsilon_0 n^2 h^2 / Z\pi me^2$

(b) State Law of Mass Action. Derive an expression of equilibrium constant Kc for general reversible reaction: $aA + bB \rightleftharpoons cC + dD$. Find equilibrium constant when 2.94 moles of Iodine and 8.1 moles of hydrogen were mixed and heated at 444°C until the equilibrium was established 5.64 moles of HI were formed: $H_2 + I_2 \rightleftharpoons 2HI$

Q4.

(a) Discuss the shapes of the following molecules with the help of hybridization and VSEPR. ☆ $BF_3$ ☆ $BeCl_2$ ☆ $NH_3$

(b) Differentiate any three of the following: ☆ Principal Quantum Number and Azimuthal Quantum Number. ☆ Bonding Molecular Orbitals and Anti Bonding Molecular Orbitals. ☆ Sigma Bond and Pi Bond. ☆ Isomorphism and Polymorphism

Q5.

(a) Explain the properties of water, based on hydrogen bonding? Write the properties of crystalline solid.

(b) Describe the types of colloids. Write their properties. A certain reaction A + B → Product gave the following results: Exp.1: [A] = $1.0 \times 10^{-6}$, [B] = $9.0 \times 10^{-6}$, Initial rate = $1.98 \times 10^{-4}$ mole/dm³.s; Exp.2: [A] = $2.0 \times 10^{-6}$, [B] = $9.0 \times 10^{-6}$, Initial rate = $3.96 \times 10^{-4}$ mole/dm³.s; Exp.3: [A] = $1.0 \times 10^{-6}$, [B] = $3.0 \times 10^{-6}$, Initial rate = $6.6 \times 10^{-5}$ mole/dm³.s. On the

basis of above results find the Rate Law and the order of reaction. —OR— Write note on any one of the following: ☆ Viscosity ☆ Surface Tension ☆ London Dispersion Forces